

Full Stack Development and Deployment of a Web-Based Speech Recognition System

Week 20
25 May - 31 May 2026

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Project Overview

- MAGOR is a web-based audio and video transcription system that converts recordings into text
- **Existing Problem:**
 - Although functional, components remain incomplete and require refinement
- **Proposed Solution:**
 - Improve robustness, usability, and AI-driven recording analysis



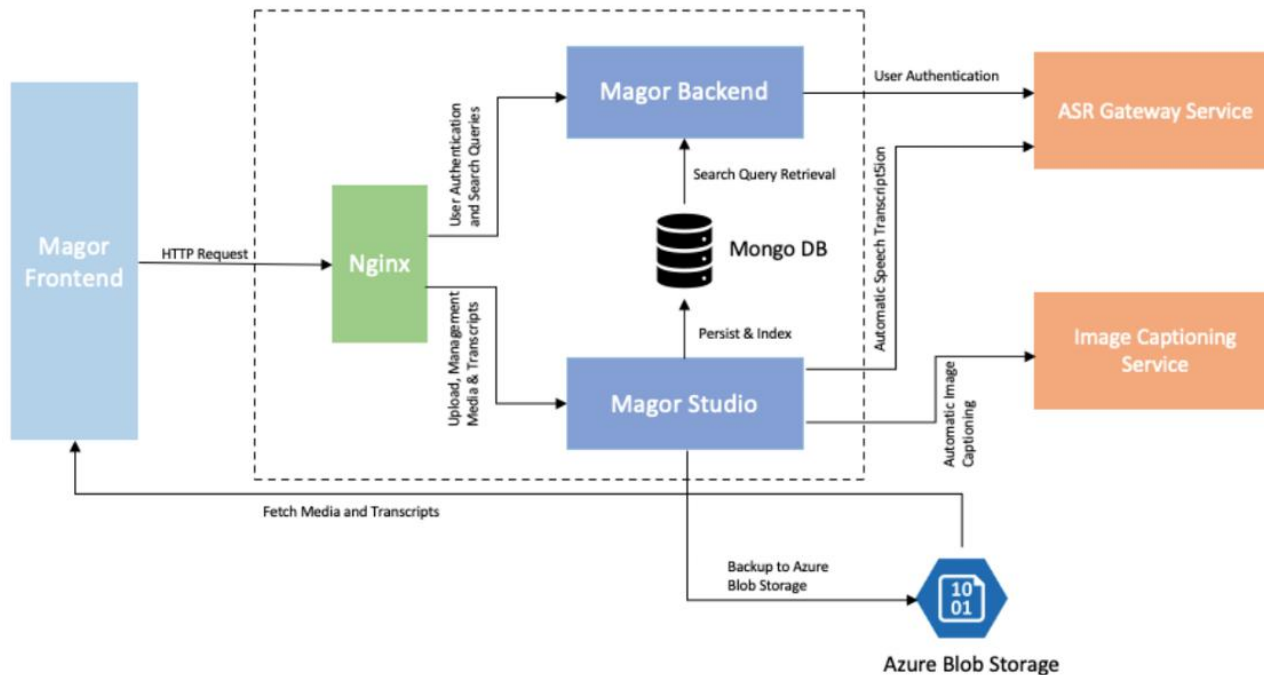
Project Objectives

The primary objectives of this project are:

- Fix system bugs and restore functionality
- Improve usability and user experience
- Enhance maintainability and scalability
- Add AI features to existing pages



Systems Overview



Source: Huang (2021)



Agenda

Work done this week:

1. Restore Admin Panel

- ASR Requests Page:
 - Automated deletion of ASR requests
- Recordings Page:
 - Automatically remove associated files when deleting recordings
- Restored Statistics Page

2. Magor Studio

- Updated transcription job submission to include speaker count



Task 1 (Fix ASR Request Deletion)

- Request status reflects deleted recordings

ASR Requests

BACK TO MAGOR

Users

recordings

ASR Requests

Statistics

Click on any field header to sort by that field

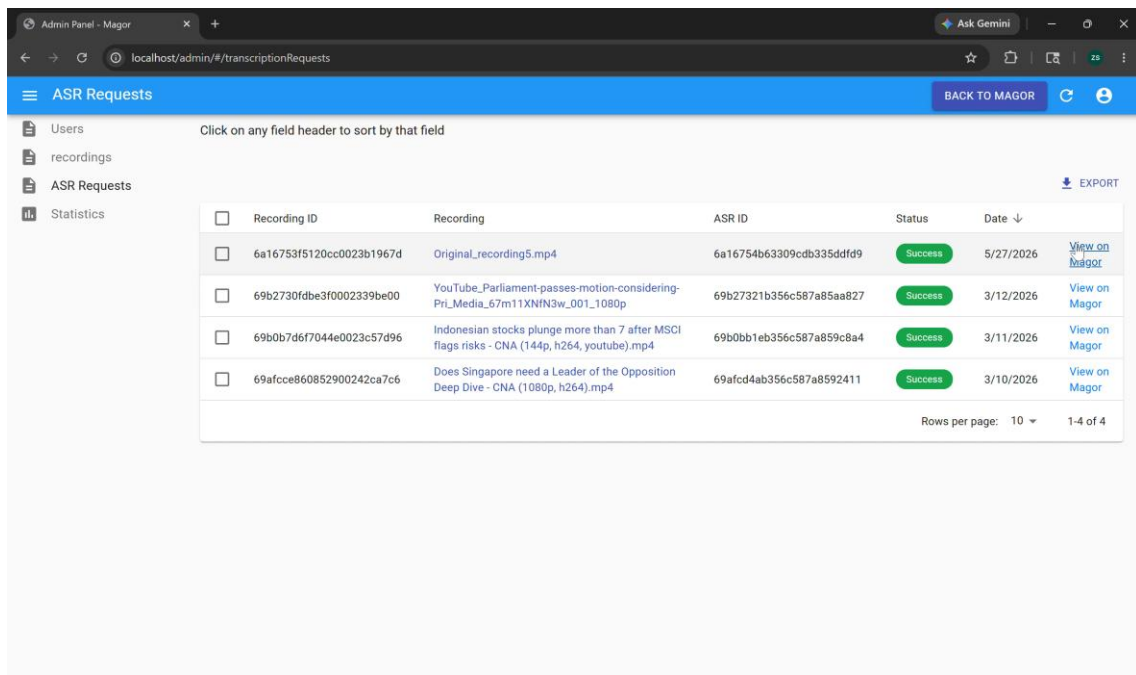
EXPORT

☐	Recording ID	Recording	ASR ID	Status	Date ↓	
☐	6a153848ea5e4a0022d9c2cc		6a15384e63309cdb335d48d8	Deleted	5/26/2026	View on Magor
☐	6a152d6714f3860023a87170	Bases Juiced, One Out.mp4	6a152d7363309cdb335d4226	Pending	5/26/2026	View on Magor
☐	69b2730fdbef0002339be00	YouTube_Parliament-passes-motion-considering-Pri_Media_67m11XNfN3w_001_1080p	69b27321b356c587a85aa827	Success	3/12/2026	View on Magor
☐	69b0b7d6f7044e0023c57d96	Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4	69b0bb1eb356c587a859c8a4	Success	3/11/2026	View on Magor
☐	69afcce860852900242ca7c6	Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4	69afcd4ab356c587a8592411	Success	3/10/2026	View on Magor

Rows per page: 10 1-5 of 5

Task 1 (Fix ASR Request Deletion)

- Updated Delete Demo:



Admin Panel - Magor

localhost/admin/#/transcriptionRequests

ASK GEMINI

BACK TO MAGOR

Users

recordings

ASR Requests

Statistics

Click on any field header to sort by that field

EXPORT

☐	Recording ID	Recording	ASR ID	Status	Date ↓	
☐	6a16753f5120cc0023b1967d	Original_recording5.mp4	6a16754b63309cdb335ddfd9	Success	5/27/2026	View on Magor
☐	69b2730fde3f0002339be00	YouTube_Parliament-passes-motion-considering-Pri_Media_67m11XNfN3w_001_1080p	69b27321b356c587a85aa827	Success	3/12/2026	View on Magor
☐	69b0b7d6f7044e0023c57d96	Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4	69b0bb1eb356c587a859c8a4	Success	3/11/2026	View on Magor
☐	69afce860852900242ca7c6	Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4	69afcd4ab356c587a8592411	Success	3/10/2026	View on Magor

Rows per page: 10 1-4 of 4



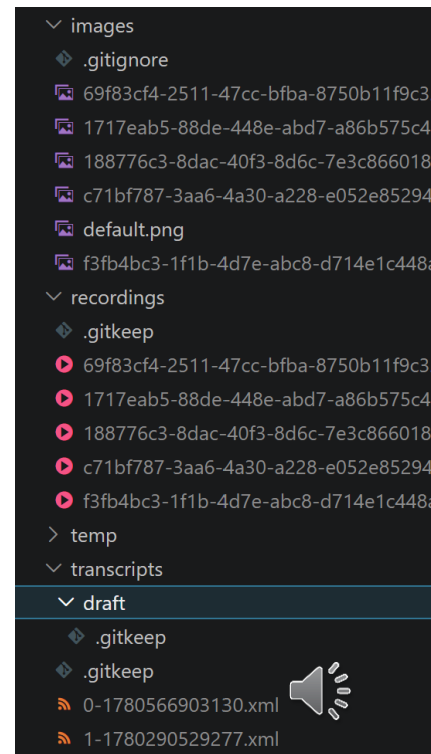
Task 2 (Update Admin Recording Deletes)

- Recordings Manager Delete button (Studio):

Delete associated files when clicked

Magor			
test			
Title	Created	Recording Type	
Original_recording5.mp4	7 days ago	video/mp4	View Edit Delete
news-video-with-captions.mp4	2 months ago	video/mp4	View Edit Delete
YouTube_Parliament-passes-motion-considering-Pri_Media_67m11XNfN3w_001_1080p	3 months ago	video/mp4	View Edit Delete
Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4	3 months ago	video/mp4	View Edit Delete
Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4	3 months ago	video/mp4	View Edit Delete

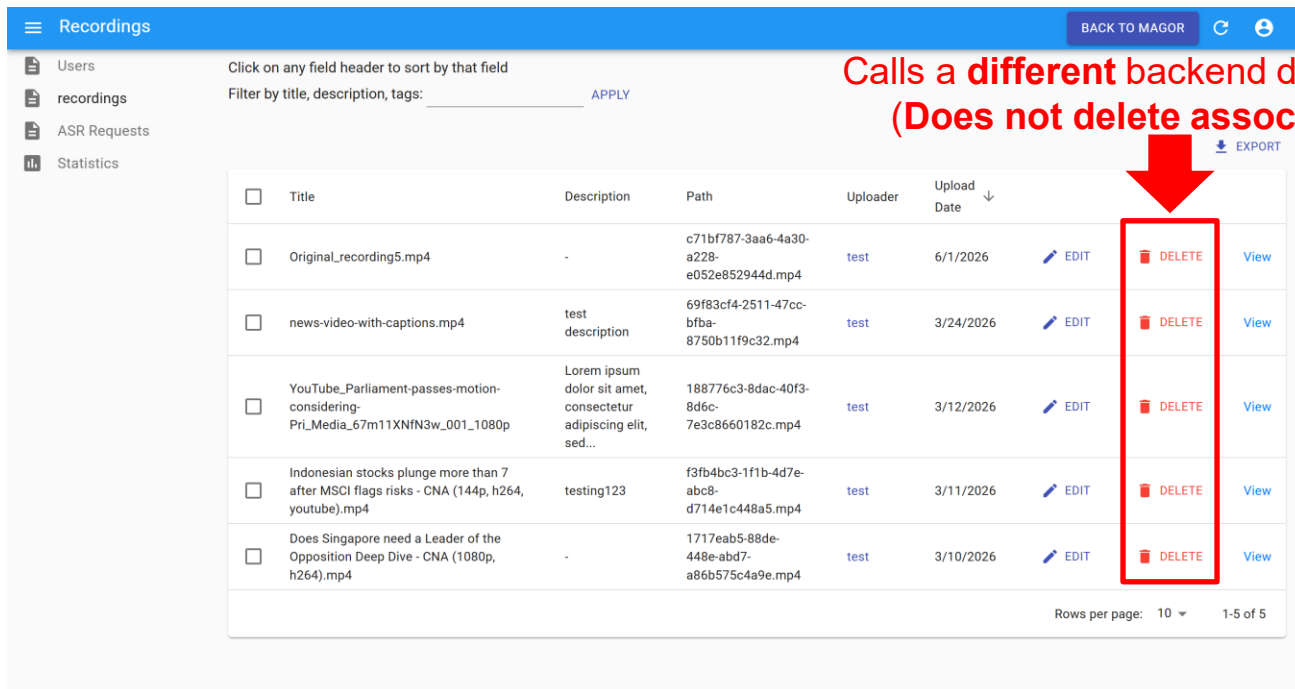
< Previous Page Showing 1 to 5 of 5 result(s) Next Page >



Associated Files

Task 2 (Update Admin Recording Deletes)

- Admin Recordings Page Delete button (Admin Panel):



The screenshot shows the 'Recordings' page in an admin panel. The page has a blue header with a menu icon, the title 'Recordings', and buttons for 'BACK TO MAGOR', a refresh icon, and a user icon. On the left, there is a sidebar with links for 'Users', 'recordings', 'ASR Requests', and 'Statistics'. The main content area has a filter bar with the text 'Click on any field header to sort by that field' and 'Filter by title, description, tags: [input] APPLY'. Below the filter bar is a table with the following columns: Title, Description, Path, Uploader, Upload Date, and actions (EDIT, DELETE, View). The table contains five rows of recordings. A red box highlights the 'DELETE' button in the first row, and a red arrow points to it from the text 'Calls a different backend delete function (Does not delete associated files)'.

☐	Title	Description	Path	Uploader	Upload Date	EDIT	DELETE	View
☐	Original_recording5.mp4	-	c71bf787-3aa6-4a30-a228-e052e852944d.mp4	test	6/1/2026	EDIT	DELETE	View
☐	news-video-with-captions.mp4	test description	69f83cf4-2511-47cc-bfba-8750b11f9c32.mp4	test	3/24/2026	EDIT	DELETE	View
☐	YouTube_Parliament-passes-motion-considering-Pri_Media_67m11XNfN3w_001_1080p	Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed...	188776c3-8dac-40f3-8d6c-7e3c8660182c.mp4	test	3/12/2026	EDIT	DELETE	View
☐	Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4	testing123	f3fb4bc3-1f1b-4d7e-abc8-d714e1c448a5.mp4	test	3/11/2026	EDIT	DELETE	View
☐	Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4	-	1717eab5-88de-448e-abd7-a86b575c4a9e.mp4	test	3/10/2026	EDIT	DELETE	View

Rows per page: 10 1-5 of 5



Task 2 (Update Admin Recording Deletes)

1. Added function to call Studio's recording deletion endpoint (used in recordingsManager)

```
113 + const deleteRecordingById = (id) =>
114 +   httpClient(`${studioUrl}/media?${stringify({ id })}`, {
115 +     method: 'DELETE',
116 +   }).then(({ json }) => ({ data: json }))
117 +
```

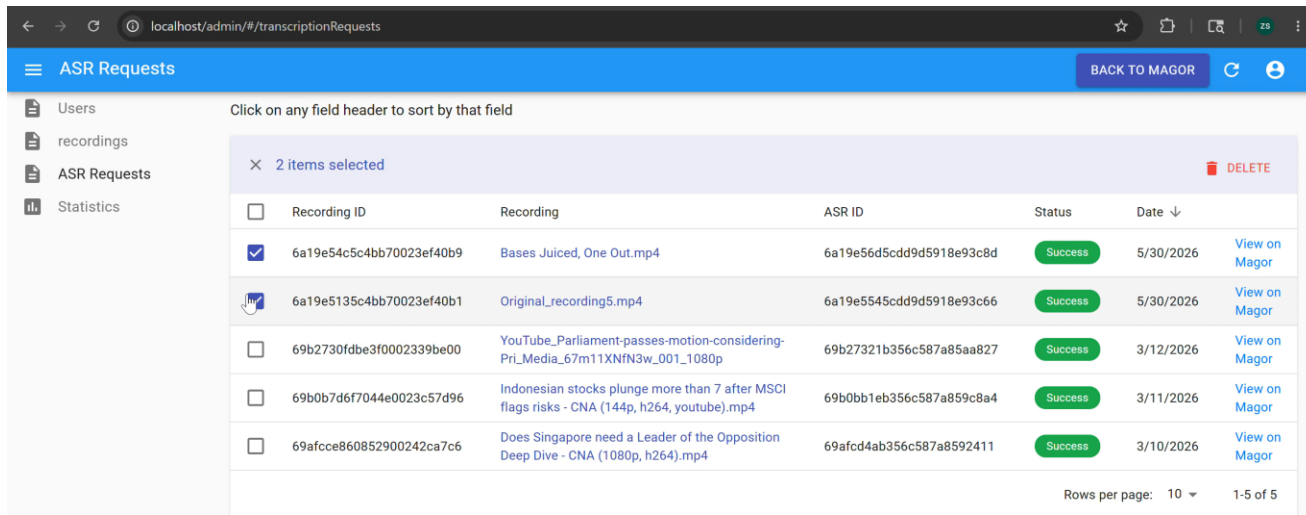
2. Frontend to redirect recording delete actions to Studio deletion workflow

```
320   delete: (resource, params) =>
321 +     resource === 'recordings'
322 +       ? deleteRecordingById(params.id)
323 +       : httpClient(`${apiUrl}/${resource}/${params.id}`, {
324 +         method: 'DELETE',
325 +       }).then(({ json }) => ({ data: json })),
326
327   deleteMany: (resource, params) => {
328 +     if (resource === 'recordings') {
329 +       return Promise.all(params.ids.map((id) =>
330 +         deleteRecordingById(id))).then(() => ({ data: params.ids }))
330 +     }
```



Task 2 (Update Admin Recording Deletes)

- Updated Admin Recording Delete Demo:



localhost/admin/#/transcriptionRequests

ASR Requests

BACK TO MAGOR

Users

recordings

ASR Requests

Statistics

Click on any field header to sort by that field

2 items selected

DELETE

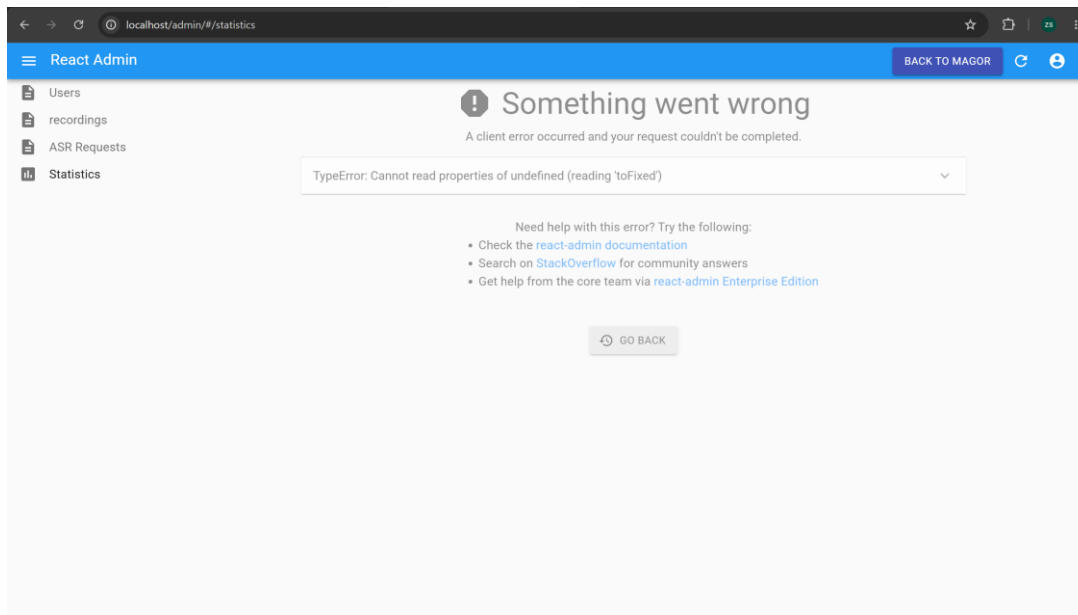
☐	Recording ID	Recording	ASR ID	Status	Date ↓	
☒	6a19e54c5c4bb70023ef40b9	Bases Juiced, One Out.mp4	6a19e56d5cdd9d5918e93c8d	Success	5/30/2026	View on Magor
☒	6a19e5135c4bb70023ef40b1	Original_recording5.mp4	6a19e5545cdd9d5918e93c66	Success	5/30/2026	View on Magor
☐	69b2730fdb3f0002339be00	YouTube_Parliament-passes-motion-considering-Pri_Media_67m11XNfN3w_001_1080p	69b27321b356c587a85aa827	Success	3/12/2026	View on Magor
☐	69b0b7d6f7044e0023c57d96	Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4	69b0bb1eb356c587a859c8a4	Success	3/11/2026	View on Magor
☐	69afce860852900242ca7c6	Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4	69afcd4ab356c587a8592411	Success	3/10/2026	View on Magor

Rows per page: 10 1-5 of 5



Task 3 (Restore Statistic Page)

- Error when loading statistics page:



Task 3 (Restore Statistic Page)

Changes to resolve Statistics Page:

1. Updated model schemas for multi-modal support and improved consistency

```
1 - const ASRRequests = require('../models/asrRequest');
2 - const Recordings = require('../models/recording');
3   const utils = require('../middleware/utils');
4
5   /**
```

```
1 + const TranscriptionRequests =
    require('../models/transcriptionRequest');
2 + const Media = require('../models/media');
3   const utils = require('../middleware/utils');
4
5   /**
```

2. Ensure consistent output by providing fallback values for empty results

```
169 +   const recordingStats = recordingsAggregation[0] ||
    emptyRecordingStats;
170 +   const mostPopularTags = await Media.aggregate([
```

```
const emptyRecordingStats = {
  count: 0,
  averageDuration: 0,
  maxDuration: 0,
  minDuration: 0,
  averageTags: 0,
  maxTags: 0,
  minTags: 0,
  averageUniqueWords: 0,
  maxUniqueWords: 0,
  minUniqueWords: 0,
  averageTranscripts: 0,
  maxTranscripts: 0,
  minTranscripts: 0,
}
```

Fallback Stats

Task 3 (Restore Statistic Page)

Changes to resolve Statistics Page:

3. Fixed incorrect field and aligned logic with updated schema

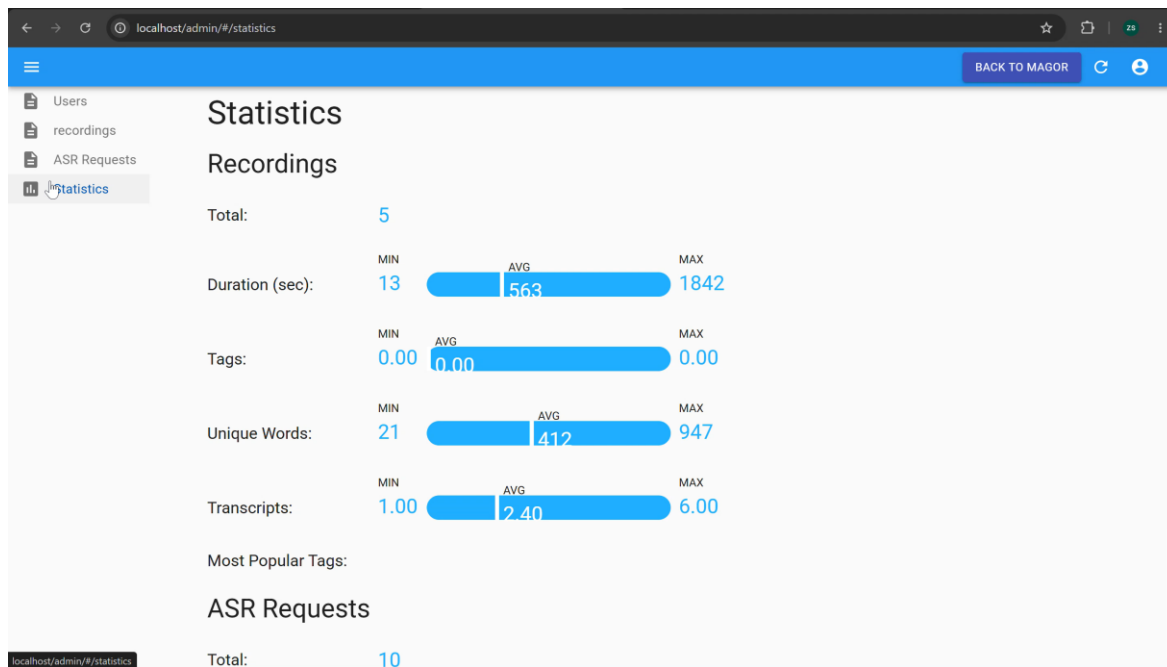
```
const oldestPendingASR = await ASRRequests.aggregate([  
  
  {$match: {completed: false}},  
  {$sort: {date: 1}},  
]);
```

```
199 + const oldestPendingASR = await  
200 +   TranscriptionRequests.aggregate([  
201 +     {$match: {status: {$ne: 'success'}}},  
202 +     {$sort: {createdAt: 1}},  
    ]);
```



Task 3 (Restore Statistic Page)

- Current statistics page demo:



Task 4 (Update Transcription Job Request Submission)

- Previous transcription job request fields:



The screenshot shows the Magor Studio interface. At the top is a blue header with 'Magor Studio' and a 'test' dropdown. Below the header, a grey box states 'No jobs running currently.' The main section is titled 'Send a new transcription job' and is enclosed in a red border. It contains three dropdown menus: 'Language' (set to 'Abax Default (This model is customised for Singapore English Mandarin Malay)'), 'Audio Type' (set to 'Close Talk'), and 'Audio Track' (set to 'Single Channel'). A blue 'Submit' button is at the bottom of this section. Below the red-bordered section is the 'Upload Transcript' section, which includes a 'Select Transcript' label and a 'Choose File' button next to the text 'No file chosen'. A small speaker icon is located in the bottom right corner of the interface.

Magor Studio test

Current Running Jobs

No jobs running currently.

Send a new transcription job

Language

Abax Default (This model is customised for Singapore English Mandarin Malay)

Audio Type

Close Talk

Audio Track

Single Channel

Submit

Upload Transcript

Select Transcript

Choose File No file chosen

Task 4 (Update Transcription Job Request Submission)

- ASR Gateway Documentation:

Submit a job

POST `https://gateway.speechlab.sg/speech`

Headers

Name	Type	Description
Authorization	string	Access Token. Bearer <access token obtained from login>

Request Body

Name	Type	Description
file	object	File Object
customData	JSON object	To pass other extra metadata such as numSpk and orgID. Example: <pre>{"numSpk":-1}</pre> -1 if no. of speaker is unknown.



Add numSpk to track speaker count



Task 4 (Update Transcription Job Request Submission)

- Current transcription job request demo:

Magor Studio

test

Current Running Jobs

No jobs running currently.

Send a new transcription job

Language

Abax Default (This model is customised for Singapore English Mandarin Malay)

Audio Type

Close Talk

Audio Track

Single Channel

Number of Speakers

Auto Detect

Submit

Upload Transcript

Select Transcript



Task 4 (Update Transcription Job Request Submission)

- POST Gateway Request Preview:

```
▼ Request Payload  View source
{
  language: "trilingual",
  audioType: "closetalk",
  audioTrack: "single",
  customData: {num_spk: 2}
}
```

Request Payload

```
X  Headers  Payload  Preview  Response  Initiator  Timing
1  {
  "success": true,
  "transRequest": {
    "status": "pending",
    "_id": "6a25a5c22bcd50023ba55b2",
    "mediaId": "6a1c7e26e8906f002330a2a3",
    "audioType": "closetalk",
    "audioTrack": "single",
    "language": "trilingual",
    "customData": {
      "numSpk": -1
    },
    "astId": "6a25a5c23949262333487360",
    "createdAt": "2026-06-07T17:09:22.846Z",
    "updatedAt": "2026-06-07T17:09:22.846Z"
  }
}
```

Mismatched
Response

(Gateway not receiving
correctly)



Response

Plans for Next Week

- Magor Studio
 - Update transcription job submission payload to ensure speaker count is correctly processed by the ASR Gateway
- Update Statistics Page to display Tag metrics correctly
- Review and plan Node.js and dependency version upgrades

